GD HW1

1) f is convex $\sim f(tx+(1-t)y) \leq t f(x) + (1-t) f(y) \ \forall x,y \ \forall t \in [0,1]$

$\Leftrightarrow f\left(\sum_{i=1}^{k} \alpha_i x_i\right) \leq \sum_{i=1}^{k} \alpha_i f(x_i) \ \forall x_1,\dots,x_k \in D \ \forall \alpha_1,\dots,\alpha_k \in [0,1] \ \sum_{i=1}^{k} \alpha_i = 1$

$\Leftarrow$) set $k=2$: $f(\alpha_1 x_1 + \alpha_2 x_2) \leq \alpha_1 f(x_1) + \alpha_2 f(x_2)$

implies f is convex $(t = \alpha_1, 1-t = \alpha_2)$

$\Rightarrow$) induction on k:

<u>base case</u> $k=2$ same as above

<u>inductive step</u> assume it holds for $k-1$

$f\left(\sum_{i=1}^{k} \alpha_i x_i\right)$

$= f\left((1-\alpha_k) \sum_{i=1}^{k-1} \frac{\alpha_i}{1-\alpha_k} x_i + \alpha_k x_k\right)$ separate x_k

$\leq (1-\alpha_k) f\left(\sum_{i=1}^{k-1} \frac{\alpha_i}{1-\alpha_k} x_i\right) + \alpha_k f(x_k)$ 2 point convexity

$\leq (1-\alpha_k) \sum_{i=1}^{k-1} \frac{\alpha_i}{1-\alpha_k} f(x_i) + \alpha_k f(x_k)$ k-1 point assumption

$= \sum_{i=1}^{k-1} \alpha_i f(x_i) + \alpha_k f(x_k)$ cancel $(1-\alpha_k)$

$= \sum_{i=1}^{k} \alpha_i f(x_i)$ □

2) $f(x,y) = x^2 + e^x + y^2 - xy$ on domain $K = [-2,2] \times [-2,2]$

prop 2.6 | $L = \max_K \|\nabla f\|$

prop 2.8 | $\beta = \max_K \lambda_{max}(\nabla^2 f)$

prop 2.11 | $\alpha = \min_K \lambda_{min}(\nabla^2 f)$

$\nabla f = \begin{bmatrix} 2x + e^x + y \\ 2y - x \end{bmatrix} \quad \|\nabla f\| = \sqrt{(2x + e^x - y)^2 + (2y-x)^2}$

since gradient is monotone in each variable the maximum is in one of the corners $(\pm 2, \pm 2)$

x	y	$\|\nabla f\|$
2	2	9.6
2	-2	14.7 $= L$
-2	2	8.4
-2	-2	2.7

$$\nabla^2 f = \begin{bmatrix} 2+e^x & -1 \\ -1 & 2 \end{bmatrix}$$

$\det(\nabla^2 f - \lambda I) = 0$ for eigenvalues

$\lambda^2 - (4+e^x)\lambda + (3+2e^x) = 0$

$$\lambda_{max} = \frac{(4+e^x) + \sqrt{e^{2x}+4}}{2}$$

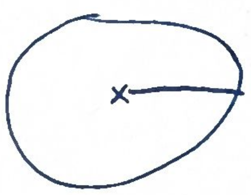
$$\lambda_{min} = \frac{(4+e^x) - \sqrt{e^{2x}+4}}{2}$$ Quadratic formula

$\beta = \max\limits_x \lambda_{max} = \underline{9.5}$ at 2

$\alpha = \min\limits_x \lambda_{min} = \underline{1.1}$ at -2 also implies (strong) convexity

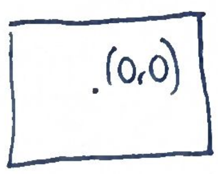
3) Projections:

- domain $x^2 + y^2 \leq 1.5$ $\quad \Pi_D(p) = \begin{cases} p & ; \|p\| \leq r \\ \sqrt{1.5}\,\frac{p}{\|p\|} & ; \|p\| > r \end{cases}$

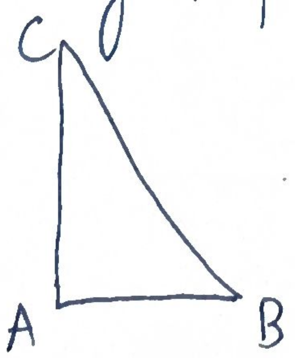


$r = \sqrt{1.5}$

- domain $[-1,1] \times [-1,1]$ $\quad \Pi_D(x,y) = (\Pi_{[-1,1]}(x), \Pi_{[-1,1]}(y))$



- domain triangle, projection defined by case analysis of 7 options



4) $f(x,y) = x^2 + 2y^2, \quad x_1 = (1,1)$

$$\nabla f = \begin{bmatrix} 2x \\ 4y \end{bmatrix} \qquad x_2 = x_1 - \gamma \nabla f(x_1) = \begin{bmatrix} 1-2\gamma \\ 1-4\gamma \end{bmatrix}$$

a) minimize $f(x_2) = (1-2\gamma)^2 + 2(1-4\gamma)^2$ "function value"

$$= 3 - 20\gamma + 36\gamma^2$$

$$\frac{d f(x_2)}{d\gamma} = -20 + 72\gamma = 0$$

$$\gamma = \frac{5}{18}$$

$$\rightarrow x_2 = \left(\frac{4}{9}, -\frac{1}{9}\right) \text{ and } f(x_2) = \underline{\underline{\frac{2}{9}}}$$

$x^* = (0,0)$

b) minimize $\|x_2\|^2 = (1-2\gamma)^2 + (1-4\gamma)^2$ "distance to minimum"

$$= 2 - 12\gamma + 20\gamma^2$$

$$\frac{d\|x_2\|^2}{d\gamma} = -12 + 40\gamma = 0$$

$$\gamma = \frac{3}{10}$$

$$\rightarrow x_2 = \left(\frac{2}{5}, -\frac{1}{5}\right) \text{ and}$$

$$\|x_2 - x^*\| = \sqrt{\left(\frac{2}{5}\right)^2 + \left(\frac{1}{5}\right)^2} = \underline{\underline{\frac{1}{\sqrt{5}}}}$$